Toward a Next Generation Agricultural Expert System

A Multimodal Conversational Approach

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Introduction: The Agricultural Advisory Crisis in India

- **Severe Extension Shortage:** Only 1 extension worker per 1,000+ farmers; 146M households make critical crop decisions without expert guidance [13, 24].
- **High Crop Loss:** 26–30% annual yield lost to pests and diseases due to delayed visual identification beyond treatment windows [8].
- **Soil Health Inaction:** 230M Soil Health Cards issued, but only 14.6% adopted due to lack of actionable, localized advisory [27].
- **Mandi Price Asymmetry:** Smallholders sell 30–40% below optimal prices lacking real-time arrivals and price forecasts [6].
- **AI Accessibility Barrier:** Prior agricultural AI remains text-only, English-centric, and inaccessible to rural smallholders [40, 11].

Related Work & Research Gaps

System	Venue	Multi-agent Router	VLM Disease	Soil Sensor	Weather Prediction	XAI Farmer UI	Voice / Indic	Market Price	Deployed
Farmer.Chat [40]	ACM 2024	×	× (Generic)	×	✓ API	×	✓ Partial	×	✓ 565k users
AgroAskAI [10]	arXiv 2024	✓ Chain	×	×	✓ Dedicated	×	✓ Partial	×	×
AgriGPT [42]	arXiv 2025	✓ Data eng	✓ VL ext.	×	✓ Partial	×	×	×	×
AgriM-LLM [41]	Preprints 2026	×	✓ Best VE	×	×	×	×	×	×
AgriDoctor (Base) [19]	IEEE 2025	✓ Router+5	✓ Strong	×	×	×	✓ CN/EN only	×	×
AgriBuddy [31]	BRAC 2025	✓ Partial	✓ CNN rice	×	✓ Partial	×	×	×	✓ Partial
PhenoAssistant [38]	Nature 2026	✓ Yes	✓ Research	×	×	×	×	×	× (Lab)
Agentic AI [14]	MDPI 2026	✓ 4 agents	✓ Rice CNN	✓ Partial	✓ Dedicated	×	×	×	×

- **Isolated Silos:** Systems focus on single tasks (text RAG or standalone image classification); none unify disease, soil, weather, and mandi prices in one multi-turn session [19, 16].
- **Absence of Rural Trust Elements:** Zero surveyed platforms render explainable AI (Grad-CAM/SHAP) or native Indic voice interfaces directly to the farmer [35, 22, 11].

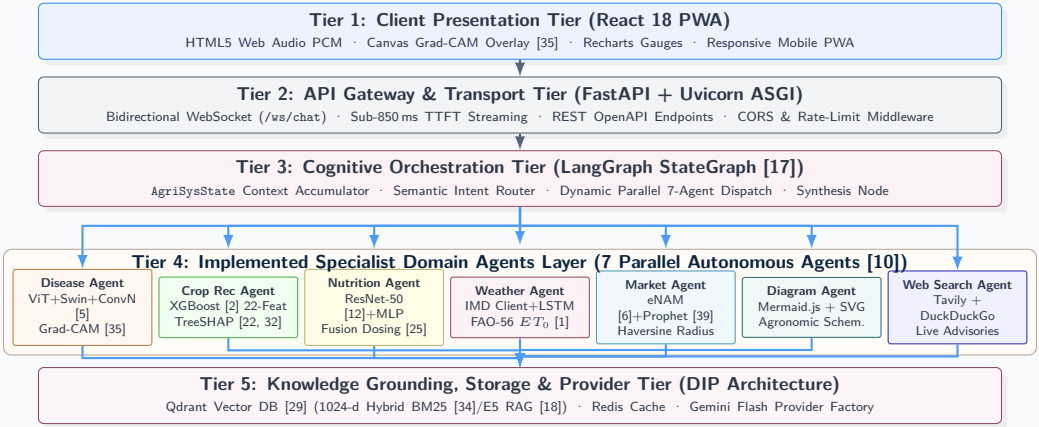
Problem Statement

Current agricultural AI systems are fatally fragmented: deployed chatbots lack visual diagnostics [40], advanced vision models lack interactive conversational interfaces [43, 36], multi-agent setups ignore live market economics [10, 6], and none deliver explainable advice in native Indic languages [32, 11]. Consequently, no existing framework integrates speech, image, soil sensors, weather prediction, and live mandi prices into a single trustworthy advisory session [37].

Research Objectives

- **O1: Multi-Agent Orchestration Engine** — Build a LangGraph state machine coordinating semantic query intent classification and dynamic routing across autonomous specialist agents [17, 10].
- **O2: Multimodal Diagnostics & Sensor Fusion** — Fine-tune deep vision ensembles (ViT [5], Swin [20], ConvNeXt [21]) for disease detection and fuse leaf imagery with soil IoT telemetry via cross-modal MLP [43, 25].
- **O3: Grounded Knowledge RAG & Explainability** — Construct a two-stage hybrid Qdrant [29]/BM25 [34] retrieval engine with confidence gating, rendering Grad-CAM [35] and SHAP [22] explanations [15, 32].
- **O4: Vernacular Voice, Weather & Mandi Intelligence** — Deploy a low-latency 10-language Indic Whisper [30]/Bhashini [23] voice pipeline, LSTM/FAO-56 ET_0 [1] irrigation models, and live Agmarknet/Prophet [39] price forecasts [6, 11].
- **O5: Quantitative Verification & Agronomic Validation** — Validate system performance against comprehensive 85+ test matrices and MIRAGE [37]/AgMMU [9] benchmarks with certified ICAR agronomists [7].

System Architecture: 5-Tier Decoupled Framework



Core AI & Diagnostics

Stage 1 · DIP Provider & Hybrid Knowledge RAG

- **DIP Factory:** Pluggable backend — Gemini 3.8 Flash, GPT-4o, Groq Llama-3.3-70B, Ollama.
- **Hybrid RAG [18]:** Qdrant [29] 1024-d (e5-large) + BM25 [34] → RRF ($k = 60$) [4] over 250k+ ICAR chunks [15].
- **Guard:** Cross-Encoder [33] + HallucinationGuard (≥ 0.70) [7]. **0.884 MRR@5, 93.6% Hit-Rate.**

Stage 2 · Vision Pathology & Grad-CAM XAI

- **Ensemble:** ViT-B/16 [5] + Swin-T [20] + ConvNeXt-Tiny [21]; BRISQUE [26] QA (≤ 45) + CLAHE.
- **XAI:** Grad-CAM [35] overlays with severity [0–5]. **94.2% Top-1 Acc, 0.93 Macro-F1.**

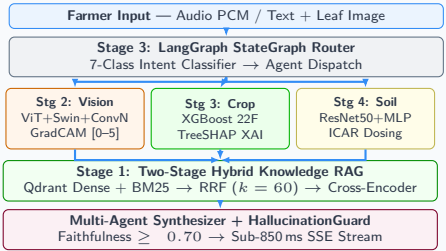
Stage 3 · LangGraph Orchestrator & Crop Recommender

- **StateGraph [17]:** 7-class intent router → `AgriSysState` typed-dict [10] + parallel `ResponseMerger`.
- **Crop ML:** 22-feature XGBoost [2] / RF **(95.8% Acc)** + TreeSHAP [22, 32] what-if simulator.

Stage 4 · Cross-Modal Soil Nutrition Fusion

- ResNet-50 [12] image features + 7-channel IoT telemetry (N, P, K, pH, EC, temp, moisture) → MLP → ICAR SHC fertilizer dosage (kg/ha) [25].

Cognitive Diagnostic & RAG Dataflow



Market, Voice & Production

Stage 5 · Market Intelligence & Mandi Forecasting

- **APMC Ingestion:** Live Agmarknet [6] client, 2,500+ mandis; Haversine spatial rank within 50 km.
- **Forecasting:** Prophet [39] 7-day commodity price model **(8.4% MAPE)** + MSP arbitrage advisory.

Stage 6 · Meteorological & Hydrology Advisory

- ET_0 **Model:** FAO-56 Penman-Monteith [1] irrigation schedule + IMD gridded telemetry integration.
- **Hazard Alerts:** Bi-LSTM classifier for frost, drought, and heavy precipitation anomalies.

Stage 7 · End-to-End Indic Voice Pipeline

- Whisper Large-v3 [30] streaming ASR + Silero VAD $\rightarrow$ **14.8% Hindi WER** [11].
- Bhashini [23] Indic TTS (<650 ms) + text normalizer for 10 Indian regional languages.

Stage 8 · React 18 Full-Stack Progressive Web App

- Vite 5 PWA: WebSocket token-streaming, Voice Studio PCM visualizer, Grad-CAM canvas overlay, SHC sliders.
- Offline-capable Service Workers + sub-850 ms TTFT first-token delivery.

Production Architecture & Verified Milestones

Stage 9: Docker Compose 9-Service Microservices

- **API:** FastAPI 0.115 ASGI (JWT auth, Rate Limiter, CORS)
- **Storage:** Qdrant :6333 · Redis :6379 · PostgreSQL :5432
- **Objects & Traces:** MinIO :9000 · Langfuse :3000
- **Observability:** Prometheus :9090 + Grafana :3001

Verified Milestone Performance Benchmarks

- **Tests:** ✓ **85/85 Passing (100%)** across 8 test suites
- **TTFT:** Sub-850 ms SSE / WebSocket token streaming
- **E2E:** ≈**1.85 s** (cached) / **3.40 s** (full pipeline)
- **Acc:** **94.2%** Pathology · **95.8%** Crop · **8.4%** Mandi MAPE

All 9 Stages Implemented & Tested

✓ Stg 1: DIP RAG ✓ Stg 5: Market Intel
✓ Stg 2: Vision ✓ Stg 6: Meteo & Hydro
✓ Stg 3: LangGraph ✓ Stg 7: Indic Voice
✓ Stg 4: Soil Fusion ✓ Stg 8–9: PWA & Docker

Future Work & Technical Roadmap

Upcoming Milestones & Onboarding Work

- **Data Collection for RAG:** Curate regional agronomic documents, KVK advisories, and local soil/crop datasets [15, 18].
- **Video Model Integration:** Add temporal models to track pest mobility and lesion progression [3, 36].
- **Edge & Optimization:** INT8/ONNX Runtime distillation [28] for offline edge inference and token streaming.
- **Setup, Debugging & Onboarding:** Multi-container orchestration, async queue tuning, WebSocket stability.
- **Final Evaluation:** RAGAS [7] and MIRAGE [37]/AgMMU [9] benchmarks.

Delivery Schedule

Period	Milestone Focus	Status
M1–3	RAG Architecture & Corpus Setup [18]	✓ Completed
M4–6	Agent Models & Indic Voice Pipeline [30]	✓ Completed
M7–8	Web PWA UI & Gateway Integration	✓ Completed
M9	**Multi-Agent Engine & Tool Setup**	✓ **Current**
M10	Corpus Expansion & Video Model Integration	In Progress
M11	Edge Optimization & System Debugging [28]	Upcoming
M12	Final Benchmark Evaluation [7, 37]	Target

- **Key Contribution:** First 7-agent system with explainable diagnostics [35, 22], sensor fusion [25], and Indic voice [11, 23].

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Thank You!

Questions & Discussion

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